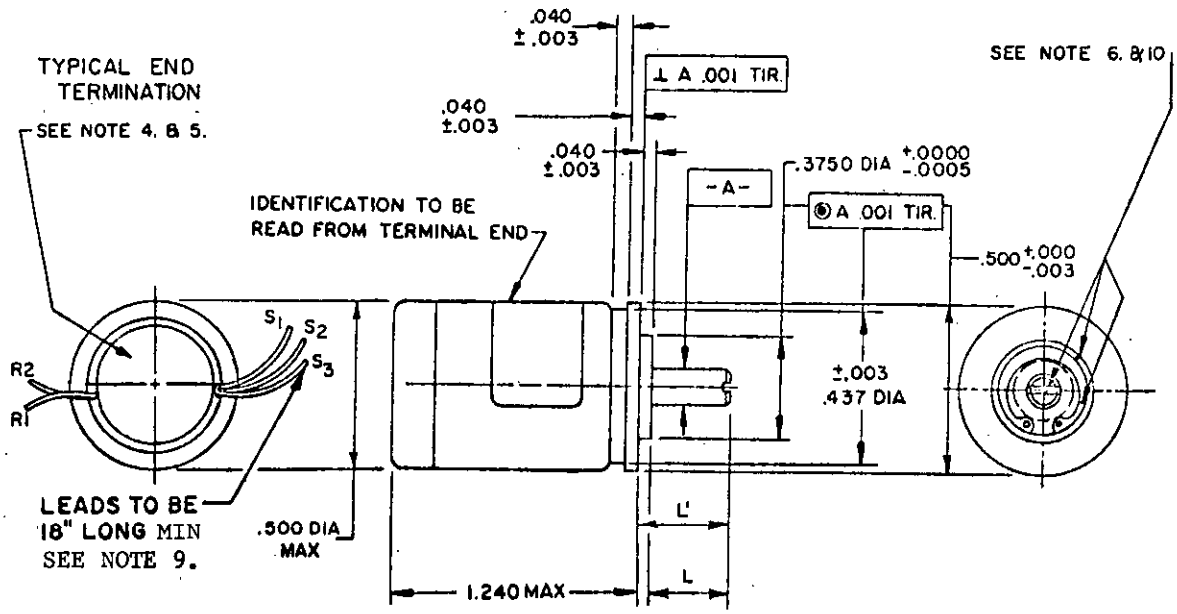
MIL-S-20708/136A
9 July 1971
SUPERSEDING
MIL-S-20708/136
12 September 1966

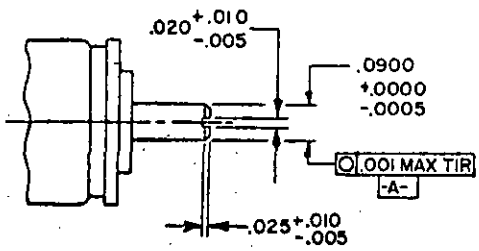
MILITARY SPECIFICATION SHEET

SYNCHRO, CONTROL TRANSFORMER, TYPE 26V-05CT4A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the synchro described herein shall consist of this document and the latest issue of Specification MIL-S-20708.





FSC 5990

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are $\pm$ .005 and angles $\pm$ 1°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal lead wires shall depart through the back surface of the terminal end of the synchro in any acceptable manner.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Mark on housing is to match shaft mark within 10° when rotor is set at approx. synchro zero, and will bear no relation to the center lines shown on the drawing.
7. End play shall be measured as total movement during reversal of 4 oz. axial load.
8. Radial play shall be measured as close to the bearing as possible during reversal of 4 oz. load applied not more than 1/8 inch from end of shaft.
9. Wire leads shall be type ET-30, MIL-W-16878/6.
10. The front retainer shall be threaded or a snap ring.
11. Pending issuance of a qualified products list (QPL) a sample of the synchros to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-S-20708.
12. Endurance shall be performed at 600 rpm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.

SYNCHRO, CONTROL TRANSFORMER, TYPE 26V-05CT4A

TABLE I PERFORMANCE REQUIREMENTS

Requirement	Value	Unit	Tolerance	Degraded Value	Degraded Value 1/
Frequency	400	Hz	±1%	--------	--------
Primary Voltage	11.8	volts	±1%	--------	--------
Primary Current	20.0	milliamps	maximum	--------	--------
Primary Power	0.045	watts	nominal	--------	--------
Impedance		ohms	R±20%+jx±20%	--------	--------
Zso	175+j615	--------	--------	--------	--------
Zrs	855+j240	--------	--------	--------	--------
Transformation Ratio	1.765	--------	±3%	--------	--------
Phase Shift	16 (lead)	degrees	maximum	--------	--------
Electrical Error 5/	10	minutes	maximum	--------	15
Null Voltage		millivolts	maximum	--------	--------
Total	60	--------	--------	90 2/ 3/	90
Fundamental	40	--------	--------	60 2/ 3/	60
Friction Torque	3	g. cm.	maximum	9 3/ 4/	9
Radial Play	0.0008	inches	maximum	0.0012 3/	0.0012
End Play	0.0001-0.0010	inches	min.-max.	0.0001-0.0015 3/	0.0001-0.0015
Brush Contact Res.	Meas. D.C. Rotor Res.	ohms	1 or 0.5%	--------	1.5 or 0.75%
Dielectric Withstand.	1.0 (peak)	milliamps	maximum	--------	1.5 (peak)
Insulation Res.	50	megohms	minimum	10 2/	25
Wire Lead Stress 6/	1	pound	maximum	--------	--------
Temperature Rise	10	°C	maximum	--------	--------

NOTES:
1/ Applicable after High Impact Shock
2/ Applicable during High Temperature
3/ Applicable after Endurance
4/ Applicable during Low Temperature
5/ An additional tolerance of ±3 minutes is allowed to compensate for mounting type eccentricity errors for Size 05 Synchros encountered during testing
6/ Wire lead stress shall be performed as a qualification test

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.

TABLE II. Dash Number and Characteristics

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				L ref	L' dim to first mounting flange surface (±.010)
-001	.0900	Plain	Wire leads	.335	.375

Part Number: M20708/136A and a dash number from Table II.

Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Frankford Arsenal, Philadelphia, Pa. 19137, ATTN: SMUFA-T1000 and information pertaining to qualification of products may be obtained from that activity.

Custodians:
Army - MU
Navy - AS
Air Force - 80

Preparing activity:
Army - MU

Project No. 5990-0278

Review activities:
Army - EL, ME, MI, SL
Navy - AS
Air Force - 17, 11
DSA - ES

User activities:
Army - AV, WC, AT
Navy - MC, OS
Air Force - None

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.